Assignment 2 - Data Analysis using R Programming

Historical Stock Analysis of Axis Bank for Revenue Optimization

Group 5

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Introduction

This analysis examines 15-minute interval stock data for Axis Bank in 2025 to identify patterns in price movement and trading activity. The goal is to evaluate whether trading volume and price behavior can provide actionable insights for financial decision-making.

1. Load the Dataset

```r
data <- read.csv("AXISBANK_15m.csv", header = TRUE)

# Convert numeric columns
data$open   <- as.numeric(data$open)
data$high   <- as.numeric(data$high)
data$low    <- as.numeric(data$low)
data$close  <- as.numeric(data$close)
data$volume <- as.numeric(data$volume)

# Create Date column
data$Date <- as.Date(substr(data$timestamp, 1, 10))

names(data)
```

```
## [1] "timestamp" "open"      "high"      "low"       "close"     "volume"
## [7] "Date"
```

```r
nrow(data)
```

```
## [1] 1548
```

2. Structure of Dataset

```r
str(data)
```

```
## 'data.frame':    1548 obs. of  7 variables:
##  $ timestamp: chr  "2025-06-04 9:30" "2025-06-04 9:45" "2025-06-05 3:45" "2025-06-05 4:00" ...
##  $ open     : num  1174 1170 1172 1168 1168 ...
##  $ high     : num  1174 1173 1172 1171 1169 ...
##  $ low      : num  1169 1170 1162 1167 1166 ...
##  $ close    : num  1170 1172 1168 1169 1166 ...
##  $ volume   : num  3602828 4305871 659767 1175636 1369788 ...
##  $ Date     : Date, format: "2025-06-04" "2025-06-04" ...
```

3. Variables in Dataset

```r
names(data)
```

```
## [1] "timestamp" "open"      "high"      "low"       "close"     "volume"
## [7] "Date"
```

4. Top 15 Rows

```r
head(data, 15)
```

```
##          timestamp   open   high    low  close  volume       Date
## 1  2025-06-04 9:30 1173.7 1173.7 1168.7 1169.5 3602828 2025-06-04
## 2  2025-06-04 9:45 1169.6 1173.0 1169.5 1171.7 4305871 2025-06-04
## 3  2025-06-05 3:45 1172.5 1172.5 1161.5 1167.5  659767 2025-06-05
## 4  2025-06-05 4:00 1167.6 1170.9 1167.0 1168.6 1175636 2025-06-05
## 5  2025-06-05 4:15 1168.5 1169.2 1165.7 1166.1 1369788 2025-06-05
## 6  2025-06-05 4:30 1165.9 1166.4 1164.0 1165.4 1566477 2025-06-05
## 7  2025-06-05 4:45 1165.3 1166.5 1164.0 1164.5 1801242 2025-06-05
## 8  2025-06-05 5:00 1164.7 1165.6 1164.0 1164.8 2010221 2025-06-05
## 9  2025-06-05 5:15 1165.0 1165.9 1162.9 1163.4 2224413 2025-06-05
## 10 2025-06-05 5:30 1163.6 1164.3 1162.7 1162.9 2463817 2025-06-05
## 11 2025-06-05 5:45 1163.0 1167.9 1162.9 1166.2 2770671 2025-06-05
## 12 2025-06-05 6:00 1165.9 1166.4 1163.2 1164.0 2976654 2025-06-05
## 13 2025-06-05 6:15 1164.1 1164.9 1163.1 1164.4 3198644 2025-06-05
## 14 2025-06-05 6:30 1164.5 1175.0 1164.4 1173.5 3647172 2025-06-05
## 15 2025-06-05 6:45 1174.0 1175.0 1169.8 1170.9 4044772 2025-06-05
```

5. User-Defined Function

```r
calc_15m_return <- function(open_price, close_price) {
  ((close_price - open_price) / open_price) * 100
}

calc_15m_return(data$open[1], data$close[1])
```

```
## [1] -0.3578427
```

6. Filter Data (2025 High Volume)

```r
filtered_data <- data[
  format(data$Date, "%Y") == "2025" &
  data$volume > mean(data$volume, na.rm = TRUE),
]

head(filtered_data, 5)
```

```
##          timestamp   open   high    low  close  volume       Date
## 1  2025-06-04 9:30 1173.7 1173.7 1168.7 1169.5 3602828 2025-06-04
## 2  2025-06-04 9:45 1169.6 1173.0 1169.5 1171.7 4305871 2025-06-04
## 14 2025-06-05 6:30 1164.5 1175.0 1164.4 1173.5 3647172 2025-06-05
## 15 2025-06-05 6:45 1174.0 1175.0 1169.8 1170.9 4044772 2025-06-05
## 16 2025-06-05 7:00 1170.7 1171.0 1166.3 1168.4 4389714 2025-06-05
```

7. Dependent & Independent Variables

The closing price is considered the dependent variable as it represents the outcome we aim to analyze and potentially predict for revenue optimization. Trading volume and opening price are treated as independent variables because they influence price movements and trading behavior.

```r
dep_df <- filtered_data[, c("Date", "close")]
ind_df <- filtered_data[, c("Date", "volume", "open")]

# Add ID for reshaping
filtered_data$id <- 1:nrow(filtered_data)

temp_wide <- filtered_data[, c("id", "Date", "close", "volume")]

long_df <- reshape(
  temp_wide,
  direction = "long",
  varying = list(c("close", "volume")),
  v.names = "Value",
  timevar = "Metric",
  times = c("close", "volume"),
  idvar = "id"
)
head(long_df, 5)
```

```
##         id       Date Metric  Value
## 1.close  1 2025-06-04  close 1169.5
## 2.close  2 2025-06-04  close 1171.7
## 3.close  3 2025-06-05  close 1173.5
## 4.close  4 2025-06-05  close 1170.9
## 5.close  5 2025-06-05  close 1168.4
```

```r
# Merge dependent + independent
new_df <- merge(dep_df, ind_df, by = "Date")

head(new_df, 5)
```

```
##         Date  close  volume   open
## 1 2025-04-04 1097.9 3330766 1095.8
## 2 2025-04-04 1097.9 3450954 1097.8
## 3 2025-04-04 1097.9 3667370 1095.9
## 4 2025-04-04 1097.9 3793008 1097.2
## 5 2025-04-04 1097.9 3927442 1096.3
```

The reshaped dataset (long format) allows comparison of multiple variables (closing price and volume) within a single column structure, which is useful for further analysis and visualization.

8. Remove Missing Values

```r
new_df <- na.omit(new_df)
```

9. Remove Duplicates

```r
new_df <- new_df[!duplicated(new_df), ]
nrow(new_df)
```

```
## [1] 8134
```

10. Reorder Rows (Descending Closing Price)

```r
ordered_df <- new_df[order(-new_df$close), ]
head(ordered_df, 5)
```

```
##            Date  close  volume   open
## 2721 2025-04-21 1237.1 5715184 1221.3
## 2722 2025-04-21 1237.1 5985333 1218.0
## 2723 2025-04-21 1237.1 6739435 1224.9
## 2724 2025-04-21 1237.1 4619244 1220.2
## 2725 2025-04-21 1237.1 5250047 1224.8
```

11. Rename Columns

```r
colnames(new_df)[colnames(new_df) == "close"] <- "Closing_Price"
colnames(new_df)[colnames(new_df) == "volume"] <- "Trading_Volume"

head(new_df, 3)
```

```
##         Date Closing_Price Trading_Volume   open
## 1 2025-04-04        1097.9        3330766 1095.8
## 2 2025-04-04        1097.9        3450954 1097.8
## 3 2025-04-04        1097.9        3667370 1095.9
```

12. Add New Variables

```r
new_df$Trading_Volume_Doubled <- new_df$Trading_Volume * 2
new_df$Total_Trade_Value <- new_df$Trading_Volume * new_df$Closing_Price

head(new_df, 3)
```

```
##         Date Closing_Price Trading_Volume   open Trading_Volume_Doubled
## 1 2025-04-04        1097.9        3330766 1095.8                6661532
## 2 2025-04-04        1097.9        3450954 1097.8                6901908
## 3 2025-04-04        1097.9        3667370 1095.9                7334740
##   Total_Trade_Value
## 1        3656847991
## 2        3788802397
## 3        4026405523
```

13. Create Training Set

```r
set.seed(123)

train_index <- sample(1:nrow(new_df), 0.7 * nrow(new_df))
train_set <- new_df[train_index, ]

nrow(train_set)
```

```
## [1] 5693
```

14. Summary Statistics

```r
summary(new_df)
```

```
##       Date             Closing_Price  Trading_Volume          open
##  Min.   :2025-04-04   Min.   :1034   Min.   : 3292286   Min.   :1034
```

```
##  1st Qu.:2025-04-16   1st Qu.:1152   1st Qu.: 4389714   1st Qu.:1152
##  Median :2025-04-25   Median :1186   Median : 5783060   Median :1186
##  Mean   :2025-05-07   Mean   :1167   Mean   : 6549864   Mean   :1167
##  3rd Qu.:2025-06-05   3rd Qu.:1214   3rd Qu.: 7886841   3rd Qu.:1214
##  Max.   :2025-07-03   Max.   :1237   Max.   :17242785   Max.   :1238
##  Trading_Volume_Doubled Total_Trade_Value
##  Min.   : 6584572       Min.   :3.441e+09
##  1st Qu.: 8779428       1st Qu.:5.142e+09
##  Median :11566120       Median :6.661e+09
##  Mean   :13099727       Mean   :7.656e+09
##  3rd Qu.:15773682       3rd Qu.:9.265e+09
##  Max.   :34485570       Max.   :2.019e+10
```

##The summary statistics indicate that the closing price is relatively stable, with a moderate range, suggesting limited volatility within the observed period.

15. Mean, Median, Mode, Range

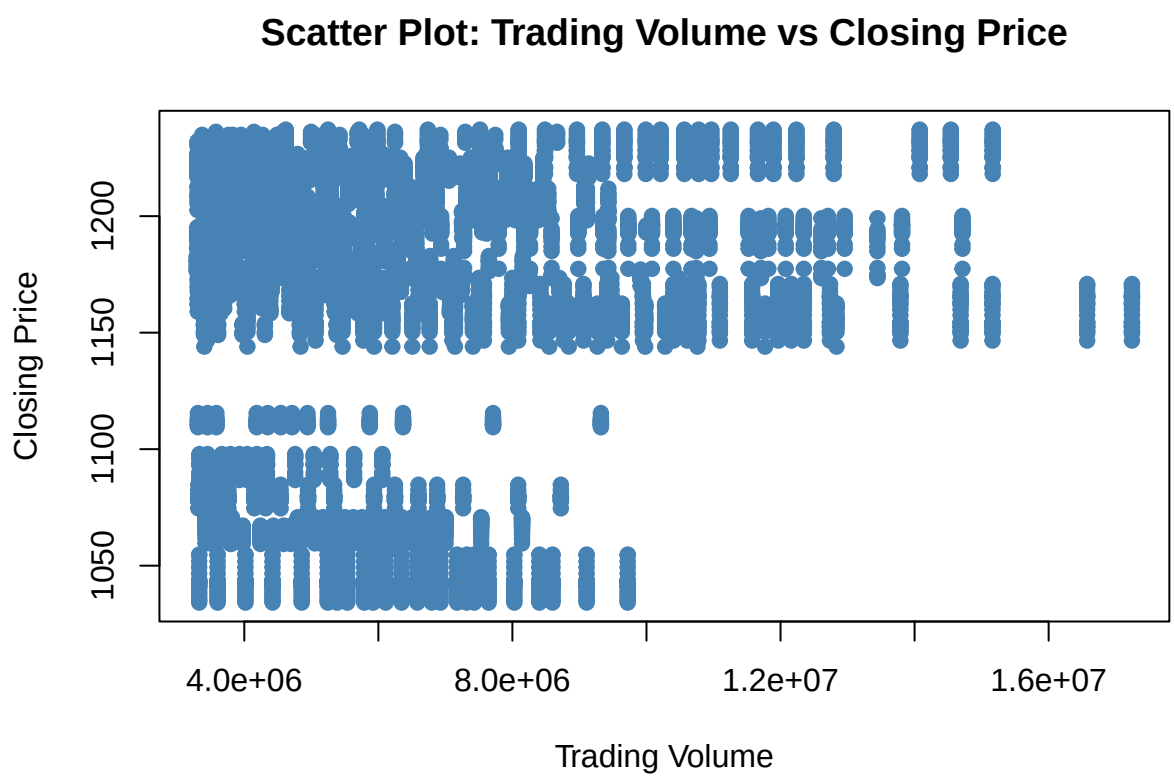
```r
mean_price   <- mean(new_df$Closing_Price, na.rm = TRUE)
median_price <- median(new_df$Closing_Price, na.rm = TRUE)

get_mode <- function(v) {
  uniqv <- unique(v)
  uniqv[which.max(tabulate(match(v, uniqv)))]
}

mode_price <- get_mode(new_df$Closing_Price)

range_price <- range(new_df$Closing_Price, na.rm = TRUE)

stats_table <- data.frame(
  Statistic = c("Mean", "Median", "Mode", "Range (Min to Max)"),
  Value = c(
    round(mean_price, 2),
    round(median_price, 2),
    round(mode_price, 2),
    paste(round(range_price[1], 2), "to", round(range_price[2], 2))
  )
)

knitr::kable(stats_table, caption = "Summary Statistics of Closing Price")
```

Table 1: Summary Statistics of Closing Price

Statistic	Value
Mean	1166.79
Median	1186
Mode	1221.1
Range (Min to Max)	1034.15 to 1237.1

16. Scatter Plot

```
plot(
  new_df$Trading_Volume,
  new_df$Closing_Price,
  main = "Scatter Plot: Trading Volume vs Closing Price",
  xlab = "Trading Volume",
  ylab = "Closing Price",
  pch = 19,
  col = "steelblue"
)
```

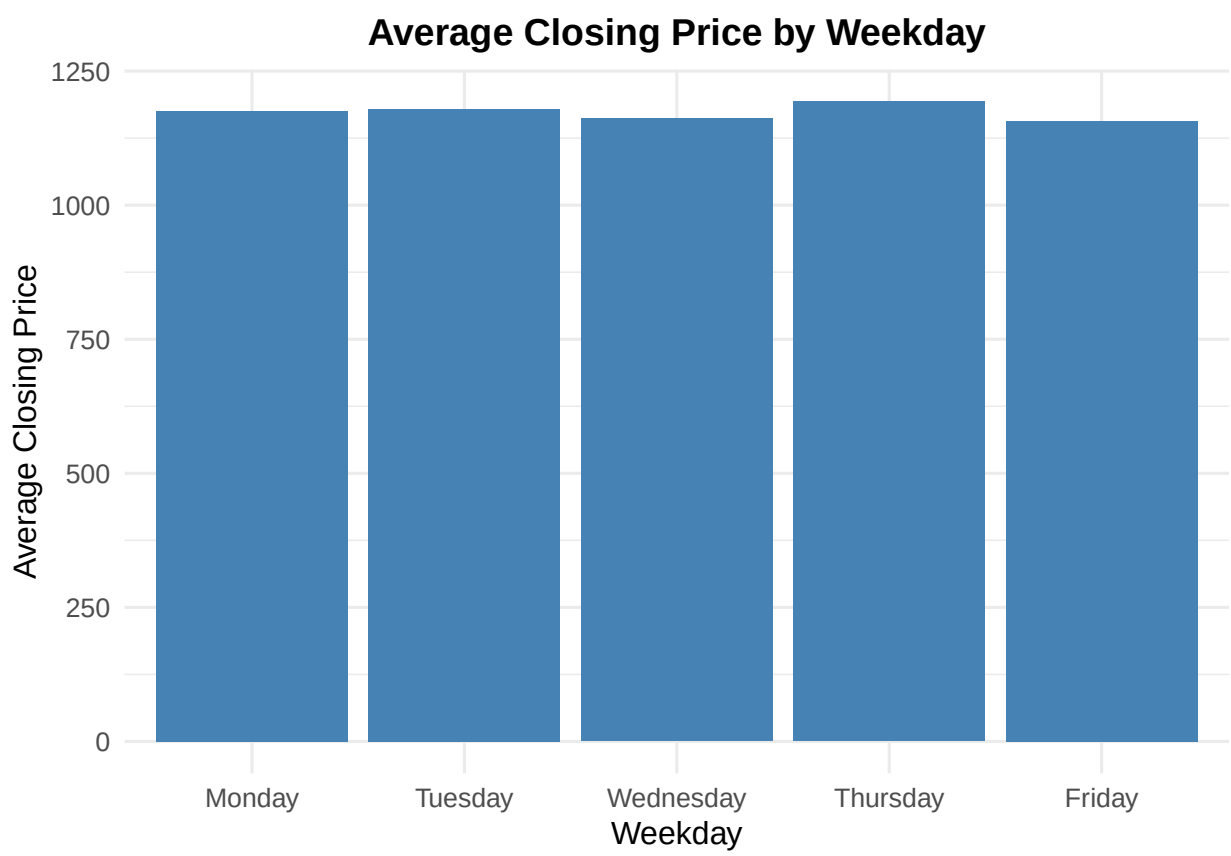


The scatter plot shows a weak and widely dispersed relationship between trading volume and closing price. This visual pattern supports the low Pearson correlation value, indicating that trading volume does not strongly influence price movement.

17. Bar Plot (Average Closing Price by Weekday)

```
library(ggplot2)
```

```r
filtered_data$Weekday <- weekdays(filtered_data$Date)

filtered_data$Weekday <- factor(
  filtered_data$Weekday,
  levels = c("Monday", "Tuesday", "Wednesday", "Thursday", "Friday")
)

avg_data <- aggregate(close ~ Weekday, data = filtered_data, mean)

ggplot(avg_data, aes(x = Weekday, y = close)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(
    title = "Average Closing Price by Weekday",
    x = "Weekday",
    y = "Average Closing Price"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.text.x = element_text(angle = 0)
  )
```



The bar plot shows variation in average closing prices across weekdays. However, no consistent trend is observed, suggesting that weekday alone is not a strong factor influencing stock price movement.

18. Pearson Correlation

```
correlation <- cor(
  new_df$Trading_Volume,
  new_df$Closing_Price,
  method = "pearson"
)

correlation
```

```
## [1] 0.08610905
```

Conclusion

The Pearson correlation value (~0.086) indicates a very weak relationship between trading volume and closing price. This suggests that trading volume alone is not a strong predictor of price movement.

While high-volume periods highlight active trading windows, additional variables such as market trends or external factors would be required for accurate revenue optimization strategies.

GitHub Repository

https://github.com/YashikaDesul/Analytics-Programming-Visualization.git